

Topic: Calculator

Hsi Chang

zhan4418@purdue.edu

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Calculator.net

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Math Calculators

Use the basic math calculator to do simple calculations or use one of the following calculators.

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Radiation Units and Conversion Factors

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Dose Unit Conversion Tool

Insert a number with up to 2 decimal points

Absorbed Dose

rad

centigray (cGy)

gray (Gy)

Dose Equivalent

rem

millisievert (mSv)

sievert (Sv)

Radioactivity Unit Conversion Tool

Insert a number with up to 2 decimal points
Results are expressed in E-notation*

curie (Ci)

becquerel (Bq)

millicurie (mCi)

megabecquerel (MBq)

*E-notation examples:
 $3.05\text{e}+9 = 3.05 \times 10^9$
 $7.26\text{e}-3 = 7.26 \times 10^{-3}$

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Exposure Unit Conversion Tool

Insert a number with up to 2 decimal points
Results are expressed in E-notation*

roentgen (R)

coulomb/kg (C/kg)

*E-notation examples:
 $3.05\text{e}+9 = 3.05 \times 10^9$
 $7.26\text{e}-3 = 7.26 \times 10^{-3}$

About 1,290,000 results

Calculator.net
https://www.calculator.net/half-life-calc...

Half-Life Calculator

Calculate the half-life, mean lifetime, and decay constant of a substance undergoing exponential decay using this online tool. Learn the definition, formula, and applications of half-life in radioactive and non-radioactive ...

$$\begin{aligned} (i) \quad \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}} &= e^{-\lambda t} = e^{-\frac{t}{\tau}} \\ (ii) \quad \ln\left(\left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}\right) &= \ln(e^{-\lambda t}) = \ln(e^{-\frac{t}{\tau}}) \\ (iii) \quad \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}} &= \frac{1}{2} \Rightarrow -\lambda t = -\ln 2 \\ (iv) \quad \ln 2 \cdot t &= t_{1/2} = \frac{1}{\lambda} \ln 2 \\ (v) \quad t_{1/2} &= \tau \ln 2 = \ln 2 \cdot \tau \end{aligned}$$

EXPLORE FURTHER

Half-life problems involving carbon-14 - ChemTeam

chemteam.info

Half-Life Calculator - Radioactive decay calculator

gigacalculator.com

Drug Half-Life Calculator

omnicalculator.com

Radioactive Decay Calculator - Free Online Calculator - BYJU'S

byjus.com

How can I calculate the half life of an element? | Socratic

socratic.org

Recommended to you based on what's popular • Feedback

Omni Calculator
https://www.omnicalculator.com/chemistry/half-life

Half-Life Calculator

Learn how to calculate half-life, the time required for half of a radioactive substance to decay, using this online tool. Enter the initial and final quantities, the time ...

Estimated Reading Time: 4 mins

Half-Life Calculator

The following tools can generate any one of the values from the other three in the half-life formula for a substance undergoing decay to decrease by half.

Half-Life Calculator

Result

half-life, $t_{1/2} = 15.051499783199$

mean lifetime, $\tau = 21.714724093908$

decay constant, $\lambda = 0.046051701862542$

quantity remains N_t	initial quantity N_0	time t	half-life $t_{1/2}$
10	100	50	
Calculate Clear			

G:\web\TG-51 TG-21 calculate.doc

Wall: Material _____ Thickness _____
 Inner diameter of chamber _____
 How was K_O determined? _____

[illegible]

Medical Physics Tools
<https://www.medical-physics-tools.org/>

TG51 second check (cylindrical chamber with pulsed beam)

Disclaimer

This online TG₅₁ spread sheet is meant as a secondary independent check of TG₅₁ measurement, or for educational purposes. It should not be used as a primary calculator for k_Q are chosen from the fit $k_Q = A + B \cdot 10^{-3} \cdot (\%dd(10)_x) + C \cdot 10^{-5} \cdot (\%dd(10)_x)^2$, given in the TG₅₁ Addendum (McEwen et al. 2014) Table I.

If a photon energy is selected, the electron specific entries are grayed out, and are highlighted with lavender if an electron energy is selected. The calculation is performed copying the data text from the I/O box below (JSON format). Pasting the text from the saved file can later be used to populate the table using the 'read' button.

P_{ion} and P_{pol} can be typed directly if those are known in advance, in which case, the reading at opposite polarity and lower voltage should not be entered.

I/O bar

[Copy table to clipboard](#)
[Read data into table](#)
[Hide needless stuff for printing](#)

Paste text from previous calc if needed, or start a new calculation

Institution

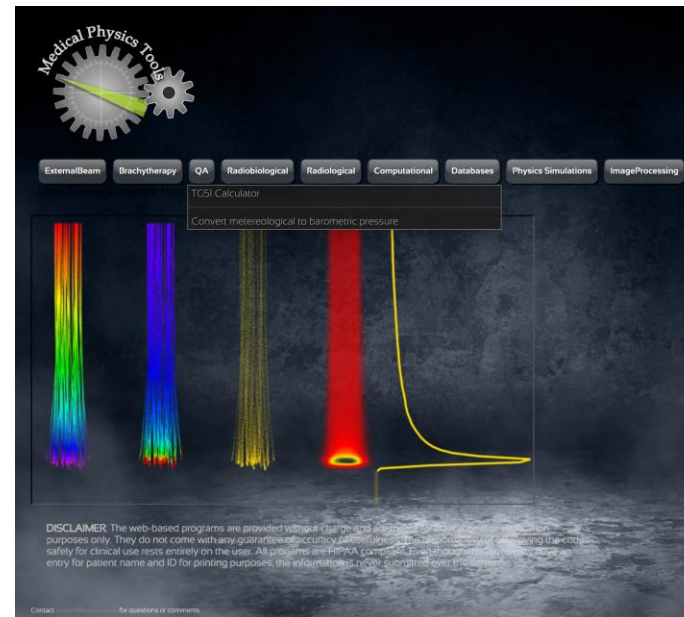
Physicist

Date

Append Column Calc

Calc

Linac	
Modality	
Energy	4MV 
Chamber model	Capintec PR-06C/G* 
Chamber s/n	
Cavity radius [cm]	
P_{rp}	
P_{detect}	
$N_{D,w}^{\text{thCo}}$ [cGy/nC]	
%dd(10) _x	
R_{90} [cm]	
$d_{\text{ref}} = 0.6R_{90} - 0.1$ [cm]	
T [°C]	
P [mmHg]	
V_H	+300V 
V_L	+150V 
M_{raw}^+ [nC]	
M_{raw}^- [nC]	
M_{raw}^H [nC]	
M_{raw}^L [nC]	
$M_{\text{raw}}(d_{\text{ref}} + 0.5r_{\text{cav}})$ [nC]	
P_{B}	
P_{TP}	
P_{pot}	
P_{ion}	
k_{ecal}	
$k'_{R_{90}}$	
k_Q	
$D_{10\text{cm}}^{\text{ph}}$ or $D_{\text{air}}^{\text{p}}$ [cGy]	
clinical TMR _{10×10(10)} or $\frac{1}{100}$ PDD(d_{ref})	
MU delivered	
$\frac{D}{\text{MU}}$ at d_{max} [cGy/MU]	



Shahid Naqvi (He/Him) · 3rd

Medical Physicist at Holy Cross Hospital Silver Spring

Baltimore, Maryland, United States · [Contact info](#)

397 connections

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 Trinity Health Cornell University

About

<https://medical-physics-tools.org>

Specialties: Medical Physics; Radiation Therapy.

Medical Linear Accelerators (Varian, Elekta, Tomotherapy),

Other radiation machines: Gamma Knife, CT simulator

HDR Brachytherapy: Accuboot, Mammosite, skin app

LDR Brachytherapy: SIRS infusion, Prostate implants, GliaSite.

EMR systems: Mosaiq, Aria.

Radiation treatment planning

Programming languages: C++/PHP/Html/CSS/Javascript/MySQL/Go

Research Interests: Radiation Physics dose calculations and Monte Carlo radiation transport.

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TG51 Protocol Calculator

N_d,w(Co-60):

1

kQ:

0.906

M_raw:

13.64nC

P_ion:

1

P_pol:

0.99

p_tp:

0.9933

p_elec:

1.001

P_leak:

0.992

p_tp:

1.1

Calculate

Result: 13.273841663073929

The equation is:

$$D_{w,Q} = N_{d,w(Co-60)} \cdot kQ \cdot M_{raw} \cdot P_{ion} \cdot P_{pol} \cdot p_{tp} \cdot p_{elec} \cdot P_{leak} \cdot p_{rp}$$

The equation with input numbers is:

$$D_{w,Q} = 1 \cdot 0.906 \cdot 13.64 \cdot 1 \cdot 0.99 \cdot 0.9933 \cdot 1.001 \cdot 0.992 \cdot 1.1$$

https://ax3301.github.io/medical_physics_dosage_calculator_24/tg51_calculator.html

Radionuclide - Radium Equivalent Msss Converter

Choose Radionuclide:

192Ir

 ▾

Activity in mCi:

15

Convert

Activity in mg-RaEq: 8.5272727272727

Activity in Bq: 3.16 \times 10^{8}

Calculation Process:

$$15 \text{ mCi} \times \frac{4.69 \frac{\text{R}\cdot\text{cm}^2}{\text{mCi}\cdot\text{h}}}{8.25 \frac{\text{R}\cdot\text{cm}^2}{\text{mg}\cdot\text{h}}} = 8.5272727272727 \text{ mg-RaEq}$$

$$8.5272727272727 \text{ mg-RaEq} \times 37 \times 10^6 \frac{\text{Bq}}{\text{mg-RaEq}} = 3.16 \times 10^8 \text{ Bq}$$

https://ax3301.github.io/medical_physics_dosage_calculator_24/radium_equivalent_msss_converter.html

